Automorphisms, forms and parabolic subgroups

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Codex, Ultra setting

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Lesson 6

Automorphisms, forms and parabolic subgroups

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A root datum determines a pinned split reductive group, but it does not remove the geometry of its subgroups or the possibility of descent. This lesson treats both. Parabolic subgroups provide the geometry needed to move Borel pairs; the pinned isomorphism theorem then determines all automorphisms; torsors record how these local descriptions descend.

Throughout, a reductive group over a scheme is smooth and affine, with connected reductive geometric fibres. A split group has a constant root datum and trivial root lines. Statements with one fixed datum are made on the corresponding open and closed part of the base. We use the preceding lessons on maximal tori, torus centralizers, root groups and the pinned isomorphism theorem. The supporting descent results are the supporting lessons Faithfully flat descent and Quotients and torsors: effective descent of affine schemes and group laws, and the classification of torsors by nonabelian first cohomology. We give the group-specific arguments here.

1. Parabolic subgroups over a field

Let first k be algebraically closed and G/k reductive. A smooth closed subgroup $P \subset G$ is **parabolic** if G/P is proper. Equivalently, it contains a Borel subgroup. Indeed, the Borel fixed-point theorem applied to the action of a Borel B on the proper quotient gives a fixed point gP, and hence $g^{-1}Bg \subset P$. Conversely, the classification below identifies every subgroup containing B with a cocharacter parabolic, whose quotient is projective.

Fix $T \subset B$, positive roots Φ^+, and simple roots Δ. For $J \subset \Delta$, choose a cocharacter λ such that

$$\langle \alpha, \lambda \rangle = 0 \quad (\alpha \in J), \qquad \langle \alpha, \lambda \rangle > 0 \quad (\alpha \in \Delta \setminus J).$$

Such a cocharacter exists by taking a rational point of the indicated face and clearing denominators. Put $P_J = P_G(\lambda)$, the subgroup on which conjugation by $\lambda(a)$ has a limit at $a = 0$. The root-coordinate construction in *Root data, Weyl chambers and the Bruhat decomposition*, Section 8, gives

$$P_J = L_J \ltimes U_J, \qquad L_J = C_G(\lambda), \tag{1.1}$$

where L_J has roots $\Phi \cap \mathbf{Z}J$, and U_J has the positive roots outside that subsystem. In particular P_J is smooth and connected and contains B.

Theorem 1.1. The parabolic subgroups containing B are exactly the P_J, for the subsets $J \subset \Delta$. Each is its own scheme-theoretic normalizer. Its quotient in G is smooth and projective.

Proof of the classification. Suppose $P \supset B$. By Bruhat decomposition, if a point of the cell Bn_wB belongs to P, then $n_w \in P$ and the entire cell belongs to P. Thus

$$W_P = \{w \in W : n_w \in P\}$$

is a subgroup of W, and P is the union of its Bruhat cells. If $w = s_{i_1} \cdots s_{i_m}$ is reduced, the closure of its cell contains the cells of each s_{i_j}. Here is the geometric reason, including the fact needed about the closure. Form the successive contracted products of the rank-one minimal parabolics P_{i_j} over B. Each rank-one quotient P_i/B is $\mathbf{P}^1$: the positive roots outside the rank-one Levi are in the shared unipotent radical, so the quotient is the rank-one Borel quotient proved in the third lesson. Their quotient by the last B is consequently an iterated $\mathbf{P}^1$-bundle. Multiplication gives a proper map to G/B. The open rank-one cells in the factors multiply isomorphically to Bn_wB/B: their conjugated roots are exactly the distinct inversion roots of the reduced word, and the ordered root-product theorem applies. The image of this proper map is therefore the cell closure. Setting all factors but the j-th to the identity includes $n_{s_{i_j}}B$ in the image. Since P/B is closed, every such simple reflection belongs to W_P.

Let $J = \{\alpha \in \Delta : s_\alpha \in W_P\}$. The preceding argument shows $W_P = W_J$, the subgroup generated by these reflections. On the other hand, $n_w \in P_G(\lambda)$ precisely when $w\lambda = \lambda$. To check this, the torus part of $\lambda(a)n_w\lambda(a)^{-1}$ is the cocharacter $w^{-1}\lambda - \lambda$; it extends to a point of the closed torus at zero only when it is zero. The stabilizer of λ in W is W_J. One can see this without a separate stabilizer theorem: if $w\lambda = \lambda$, choose a simple inversion δ of a nonidentity w. Dominance of λ and negativity of $w\delta$ force $\langle \delta, \lambda \rangle = 0$. Hence $\delta \in J$, and multiplying by s_δ lowers the length while preserving λ. Induction proves the assertion. Consequently P and P_J have the same geometric points. They are reduced closed subgroups, so they are equal.

For the normalizer, let $g \in N_G(P)(k)$. The Borels B and gBg^{-1} of the connected smooth affine group P are conjugate in P, by the field conjugacy theorem. After multiplication by an element of P, g normalizes B, so belongs to $B \subset P$. This proves equality on geometric points. It also proves equality as group schemes: the infinitesimal quotient $\operatorname{Lie} N_G(P)/\operatorname{Lie} P$ consists of P-fixed vectors in $\mathfrak{g}/\mathfrak{p}$: infinitesimal normalization gives $\operatorname{Ad}(p)X - X \in \mathfrak{p}$ for every test-valued $p \in P$. Its invariants under λ are zero, since all the weights in that quotient are strictly negative. Thus the tangent spaces of the normalizer and P agree. Its reduced group is P, and its local dimension and tangent dimension at the identity agree; it is smooth there and, by translation, everywhere. Hence $N_G(P) = P$.

Finally G/P_J is proper: the map $G/B \to G/P_J$ is surjective, and G/B is proper by the field argument in the first lesson. Apply the quotient and projectivity lemma in *Roots and reductive groups of rank one*, Section 5: a smooth connected self-normalizing subgroup with complete homogeneous fibres has a scheme quotient, and the character $\det(\operatorname{Lie} P_J)^{-1}$ defines an ample line bundle. This proves projectivity. Smoothness follows from the smooth P_J-torsor $G \to G/P_J$. □

The ordering is $P_J \subset P_K$ when $J \subset K$. The empty subset gives B; the full subset gives G. Thus allowing G itself as a parabolic adds one point to the parameter scheme.

2. Parabolics and their parameter scheme over a base

Over S, it is useful initially to allow a smooth affine monomorphism $P \to G$ whose geometric fibres are parabolic subgroups. The theorem below shows that this is automatically a closed immersion. This definition therefore does not hide a closedness hypothesis in the conclusion.

Theorem 2.1. A parabolic subgroup of a reductive S-group is, étale locally on S, a standard

cocharacter parabolic. It is closed, has connected fibres, and satisfies $N_G(P) = P$. The functor of parabolic subgroups is represented by a smooth projective S-scheme $\mathrm{Par}(G)$, and formation of this scheme commutes with every base change.

Proof of the local description. Apply the torus-lifting and conjugacy argument of the first lesson to the smooth affine group P. Étale locally, a maximal torus of a chosen geometric fibre lifts to a torus $T \subset P$. It is maximal also in G, by the field classification. Shrink and split T, and make the root datum and root lines constant as in the third and fourth lessons.

The differential of the monomorphism $P \to G$ is a fibrewise injection of vector bundles; hence its image is a locally direct summand of $\mathrm{Lie}\, G$. As a T-module it contains $\mathrm{Lie}\, T$ and either the whole of each root line or none of it. The selection is locally constant. Choose λ from the parabolic pattern in the chosen fibre. All weights of $\mathrm{Lie}\, P$ have nonnegative pairing with λ. The dynamic subgroup $P_P(\lambda) \subset P$ is smooth and has exactly those nonnegative tangent weights, by the graded-coordinate proof in the third lesson. Its inclusion is therefore étale; on every geometric fibre it is an isomorphism, since that fibre is the connected parabolic with this pattern. It follows that $P_P(\lambda) = P$. The limit functor now factors $P \to G$ through $P_G(\lambda)$. This is a monomorphism between smooth schemes, with an isomorphism on every geometric fibre; its differential is an isomorphism, so it is an étale, surjective monomorphism, and hence an isomorphism. In particular the original $P \to G$ is closed.

The formal separation and normalizer lemmas in the rank-one lesson show that $N_G(P)$ is closed of finite presentation and commutes with every base change. Their proof uses a finite prime filtration, Artin–Rees and the determinant trick to detect subgroup equations over a universal Noetherian coefficient ring; it therefore also applies on test schemes with nilpotents. Its geometric fibres are the normalizers calculated in Theorem 1.1.

We use the flat-source criterion proved in the rank-one lesson directly over the original base. Locally a finitely presented closed immersion $X \hookrightarrow Y$ over $\mathrm{Spec}\, R$ has a finitely generated ideal $I \subset C = \mathcal{O}(Y)$. If C/I is R-flat, tensoring $0 \to I \to C \to C/I \to 0$ with $\kappa(s)$ makes $I \otimes_R \kappa(s) \to C \otimes_R \kappa(s)$ injective. Equality of every geometric fibre makes this ideal zero, first after a faithful field extension and then over $\kappa(s)$. For a prime $x \subset C$ over s, this says $I_x = \mathfrak{p}_s I_x$; the finitely generated C_x-module I_x vanishes by Nakayama. Hence $I = 0$. Apply this to $P \hookrightarrow N_G(P)$, using the fibre equality in Theorem 1.1 and flatness of P/S. Neither Noetherianity of the base nor flatness of the normalizer is required.

The quotient G/P is therefore smooth and projective by the quotient lemma just used, now over S. Its ample line bundle comes canonically from $\det(\mathrm{Lie}\, P)^{-1}$.

For a pinned split G of constant datum, we obtain

$$\mathrm{Par}(G) = \coprod_{J \subset \Delta} G/P_J. \tag{2.1}$$

Here a morphism to the disjoint union may choose different J's on open and closed parts. To see that (2.1) represents the subgroup functor on every test scheme, a section of G/P_J gives a conjugate of P_J after a local lift to G. Conversely the preceding local classification supplies such lifts for any parabolic. Two lifts give the same subgroup exactly when their quotient lies in $N_G(P_J) = P_J$, so the sections glue uniquely. This proves the functorial assertion, including uniqueness over bases with nilpotents.

For an arbitrary reductive G, the local schemes just constructed have unique identifications on overlaps, because they represent the same subgroup functor. They satisfy the cocycle condition. The universal subgroup $\mathcal{P}$ gives the canonical line bundle $L = \det(\mathrm{Lie}\, \mathcal{P})^*$; its identifications also satisfy the cocycle. We show directly that these projective schemes with their line bundle descend to a projective scheme.

Work on a quasi-compact affine part of S and take finitely many affine étale charts $S_i \to S$ carrying the split descriptions. Their disjoint union is a faithfully flat quasi-compact cover. On each

chart the Grassmannian construction of AG-RG-03 Lemma 5.A, applied in its Lemma 5.1, gives a positive power of L_i and a specific closed Plücker embedding on every parabolic component. Choose a common exponent and place these finitely many components in disjoint coordinate blocks of one projective bundle, using the direct sum of their embedding modules. This gives a closed projective embedding of their disjoint union. Descend the charts, projective schemes, these specified embeddings, line bundles and their identifications to a common Noetherian coefficient stage. S02 Proposition 3.2 retains smoothness and the line bundles, and finite-presentation descent retains the closed embeddings and their finitely many identities. Thus ampleness on the model is witnessed by retained embeddings, rather than inferred from fibres of the original base.

Choose a common multiple $b > 0$ of the finitely many embedding exponents. Its sufficiently large multiples m are still very ample, by the polynomial Veronese maps. S01 Theorem 8.5 makes the positive cohomology of L_i^m vanish over each affine Noetherian model chart for sufficiently large m. To pass to all geometric fibres, use the universal finite projective complex of S02 Lemma 5.1, in degrees $[0, N]$, for this smooth projective family. Its positive cohomology is zero over that affine chart. Starting at the last term, its exact positive tail splits because the last nonzero term is projective; repeat to remove the whole positive tail. The remaining degree-zero module is finite projective, and the splitting survives every tensor. Thus $H^q(X_{i,\bar{s}}, L_i^m) = 0$ for every $q > 0$ on every geometric model fibre. A single sufficiently large common multiple works on the finitely many charts. Pull this choice back to the original cover. S02 Proposition 5.2 gives $E_i = f_{i*}L_i^m$ finite locally free, with every base change. The overlap identifications of schemes and L_i therefore identify the E_i compatibly. Faithfully flat module descent produces a finite locally free E/S.

Evaluation defines compatible closed immersions $X_i \hookrightarrow \mathbf{P}(E)|_{S_i}$. Their complete section systems contain the sections of the chosen very ample systems, so on each generating target chart the coordinate-ring map is still surjective. This first gives an immersion; the map is also proper by the graph argument of S01 Lemma 8.3, since X_i/S_i is projective. Its image is closed, and the chartwise closed immersions together with the empty inverse image off that closed image make the immersion closed globally. On affine charts of $\mathbf{P}(E)$, the ideal sheaves of these immersions have compatible faithfully flat descent data. Module descent gives their ideal submodules, and multiplication preserves them because that fact is checked on the faithfully flat cover. Their quotient algebras define a closed S-subscheme $X \subset \mathbf{P}(E)$, whose pullback is each X_i. The represented subgroup functor and smoothness descend through the same cover, so $X = \mathrm{Par}(G)$ is smooth and projective. The construction is independent of charts by its representing identity and hence glues over S. Finally the subgroup functor itself commutes with every base change; Yoneda identifies the pullback of its representing scheme with $\mathrm{Par}(G_{S'})$. □

For a nonsplit group, subsets of a chosen simple-root set need not be globally labelled. The finite étale **type scheme** is the descent of the finite set of subsets of Δ. The map from $\mathrm{Par}(G)$ to this scheme remembers the type. Borel subgroups are its component of minimal parabolics, corresponding locally to $J = \varnothing$.

3. Automorphisms of a pinned split group

Fix a pinned split group

$$(G, T, B, (E_\alpha)_{\alpha\in\Delta})$$

of based root datum $\mathcal{R}_b$. Let $\Gamma = \mathrm{Aut}(\mathcal{R}_b)$, including its action on the full character and cocharacter lattices. Regard it as a constant group scheme. The pinned isomorphism theorem gives a unique pinned automorphism s_γ inducing each $\gamma \in \Gamma$; uniqueness gives $s_\gamma s_\delta = s_{\gamma\delta}$.

Write $G^{\mathrm{ad}} = G/Z(G)$. This is the adjoint semisimple quotient: central torus directions disappear, but they remain present in Γ.

Theorem 3.1. On all S-schemes, including nonreduced ones, the automorphism functor is represented by

$$\mathrm{Aut}(G) \simeq G^{\mathrm{ad}} \rtimes \Gamma_S, \qquad (g,\gamma) \longmapsto \mathrm{Int}(g)s_\gamma. \tag{3.1}$$

This group scheme is smooth, separated and locally of finite presentation. It need not be of finite type.

Proof. Put $q : G \to G^{\mathrm{ad}}$ and $T^{\mathrm{ad}} = q(T)$. The scheme-centre calculation in RG03 Section 1 gives $Z(G) = D_S(X/\mathbf{Z}\Phi)$. The central quotient theorem of RG03 Theorem 1.1 and RG04 Theorem 7.1 therefore gives the fppf quotient q, the torus $T^{\mathrm{ad}} = D_S(\mathbf{Z}\Phi)$ and the images of all root parameters with their unchanged frames. In particular $\mathbf{Z}\Phi = \bigoplus_{\alpha\in\Delta} \mathbf{Z}\alpha$ is its character lattice. Conjugation by a section of G^{ad} is well defined on G: lift the section through q fppf locally, conjugate there, and descend the group morphisms by S04 Lemma A1.5. Two lifts differ by a section of the scheme centre and give the same morphism on every test scheme. Each s_γ preserves this centre and induces s_γ^{ad} on the quotient. The morphism in (3.1) respects multiplication, since

$$\big(\mathrm{Int}(g)s_\gamma\big)\big(\mathrm{Int}(h)s_\delta\big) = \mathrm{Int}\big(g\, s_\gamma^{\mathrm{ad}}(h)\big)s_{\gamma\delta}.$$

Let a be an automorphism after any base change $S' \to S$. Theorem 2.1 identifies the Borel aB with a section of G/B. Pulling back the fppf B-torsor $G \to G/B$ permits us locally to conjugate aB to B. Its transported aT is a maximal torus of B. The torus transporter in this smooth affine B is smooth by RG01 Lemma 3.1; its geometric fibres are nonempty by the solvable-group torus-conjugacy proof in RG01 Section 2. It is consequently a smooth surjection, and its own tautological section supplies an fppf cover on which a further conjugation in B takes this torus to T. We have reduced to an automorphism b preserving both T and B.

RG01 Lemma D.1 identifies $b|_T$ with a locally constant automorphism of the full integral character lattice, including its central directions. Preservation of B and the canonical root and coroot maps identifies its value with a section γ of $\Gamma_{S'}$. A locally constant value may be treated on its open-and-closed fibres, without assuming that there are finitely many such values. Set $\beta = s_\gamma^{-1}b$. This automorphism induces the identity on the entire based datum. On each framed simple root line it has the form $d\beta(E_\alpha) = u_\alpha E_\alpha$, with $u_\alpha \in \mathcal{O}_{S'}^\times$: an automorphism of a free rank-one module is multiplication by a unit. Because the simple roots are a basis of $X^*(T^{\mathrm{ad}})$, the rule $\alpha(t) = u_\alpha^{-1}$ defines one and only one section t of T^{ad}. Conjugation by it has precisely these multipliers on the root frames, by the quotient root-equivariance formula of RG04 Theorem 7.1. Thus $\mathrm{Int}(t)\beta$ fixes the pinning and induces the identity on its full based datum. RG05 Theorem 5.1 makes it the identity automorphism. Undoing the conjugations and using the displayed multiplication identity writes $a = \mathrm{Int}(g)s_\gamma$ fppf locally. This proves local surjectivity.

We prove injectivity on the same arbitrary test scheme. Suppose $\mathrm{Int}(h)s_\gamma = 1$. Then $\mathrm{Int}(h) = s_\gamma^{-1}$ preserves T, B and therefore their quotient torus and Borel in G^{ad}. RG04 Theorem 4.2, with the pinned representatives, identifies the entire normalizer of T^{ad} with the copies $n_w T^{\mathrm{ad}}$ labelled by the constant Weyl sheaf. Hence h has a locally constant Weyl label w. Preservation of the Borel preserves its positive root lines, so this label preserves the positive chamber. RG04 Lemma 3.A gives $w = 1$. The identity-labelled copy is the torus itself as a scheme; thus $h \in T^{\mathrm{ad}}(S')$, including when S' has nilpotents. Its inner automorphism is the identity on T, so $\gamma = 1$ on the entire datum. The equation now says that every simple frame is fixed, and therefore $\alpha(h) = 1$ for every $\alpha \in \Delta$. Their character-basis property gives $h = 1$. The kernel is trivial on every test scheme, which proves injectivity of the group-functor map.

The locally constructed factors of a consequently agree on double overlaps. S04 Lemma A1.5 glues them uniquely as sections of G^{ad} and Γ_S, and the same lemma descends their equality with a. Every construction commutes with base change, so these sections give the inverse natural transformation to (3.1). The representing scheme is the disjoint union, indexed by Γ, of copies of the smooth affine

finitely presented G^{ad}. It is smooth and locally of finite presentation. Its relative diagonal is closed: on each pair of indexed components it is either the closed diagonal of G^{ad} or empty. Thus it is separated as well. This proves all the assertions. □

For a semisimple datum, Γ is finite. For a torus of rank r, $G^{\mathrm{ad}} = 1$ and $\Gamma = \mathrm{GL}_r(\mathbf{Z})$, which is infinite for $r \geq 2$. Thus the finite-type assertion cannot be added in general.

Even a small central direction matters. On GL_2,

$$g \longmapsto \det(g)^{-1} g \tag{3.2}$$

is an involutive automorphism. On the diagonal torus it sends (t_1, t_2) to (t_2^{-1}, t_1^{-1}). It preserves the positive root and reverses the scalar direction, so is visible in the full based datum. It is not inner, since it acts nontrivially on the centre.

4. Forms and torsors

A **form** of G_0/S is an S-group which becomes isomorphic to G_0 on an étale cover. Let $A = \mathrm{Aut}(G_0)$.

Proposition 4.1. Isomorphism classes of forms of G_0 are in natural bijection with $H^1_{\mathrm{et}}(S, A)$.

Proof. Choose local isomorphisms $\phi_i : G_0|_{S_i} \to G|_{S_i}$. On overlaps,

$$a_{ij} = \phi_i^{-1}\phi_j, \qquad a_{ij}a_{jk} = a_{ik}.$$

Changing the ϕ_i's changes this cocycle by a coboundary. Equivalently, $\mathrm{Isom}(G_0, G)$ is a right A-torsor: after choosing a local isomorphism, every other one is its composite with a unique automorphism of G_0. Conversely an A-torsor has local sections on an étale cover; their transition automorphisms give a descent datum on the affine scheme G_0. S04, Lemmas A1.2 and A1.5, give effective descent of its coordinate algebras and compatible morphisms. The group operations descend with these morphisms, and their identities follow by faithful descent. Thus one obtains an affine S-group with the prescribed local isomorphisms. Its finite presentation and smoothness descend by S04, Lemma A1.3a.

For the remaining fibre assertion, take a geometric point $\mathrm{Spec}\,\Omega \to S$, with Ω algebraically closed. The pulled-back étale cover is nonempty. A nonempty affine chart of it is of finite type over Ω, so a maximal ideal gives an Ω-point by AG-CA-06, Theorem 2.1, whose field-algebra proof is Theorem 1.3 there. Pulling a covering isomorphism back along that point identifies the descended geometric fibre with $(G_0)_\Omega$. Hence that fibre is connected and reductive. This proves reductivity over every geometric point of the arbitrary base.

The two constructions recover each other's cocycles and local isomorphisms. Compatibility and uniqueness in S04, Lemma A1.5, identify their descended objects and morphisms, so they are mutually inverse on isomorphism classes. This is the concrete torsor argument from Quotients and torsors, applied to the automorphism sheaf A. □

For split pinned G_0, an **inner form with inner structure** is obtained from a G_0^{ad}-torsor by conjugation. These structures are classified by $H^1(S, G_0^{\mathrm{ad}})$. Forgetting the inner structure need not be injective on isomorphism classes. The exact sequence split by the pinning is

$$1 \longrightarrow G_0^{\mathrm{ad}} \longrightarrow A \longrightarrow \Gamma_S \longrightarrow 1. \tag{4.1}$$

The fibre over the neutral element of $H^1(S, \Gamma_S)$ is the set of $\Gamma(S)$-orbits on $H^1(S, G_0^{\mathrm{ad}})$. Indeed a trivialization of the induced Γ-torsor reduces the original cocycle to G_0^{ad}; changing that trivialization by γ changes the inner cocycle by its γ-action. This also explains why a marked inner structure and the underlying group are different classification problems.

5. Quasi-split forms

A reductive group is **quasi-split** if it admits a Borel subgroup over the base. Twisting the pinned G_0 by a Γ-torsor gives a quasi-split form: the pinned automorphisms preserve T, B and the simple-root frames as an indexed collection, so these data descend.

Over a field, this accounts for every quasi-split form of G_0, up to the corresponding based-datum form. We supply the cohomological step because the torus which occurs in it is usually twisted.

Proposition 5.1. For a field k, the forms of G_0 that admit a k-Borel are exactly the forms obtained by twisting its pinning by Γ-torsors. Their based-datum classes lie in $H^1(k, \Gamma)$.

Proof. A form with a Borel reduces its descent cocycle to the stabilizer of B in A, namely

$$\mathrm{Aut}(G_0, B) = B_0^{\mathrm{ad}} \rtimes \Gamma.$$

Fix its Γ-torsor. Relative to the canonical pinned twist, the possible reductions are torsors under the corresponding twisted B_0^{ad}. Its torus has character lattice freely based by Δ, with Galois action permuting that basis. The finite orbits of this action give finite separable fields k_i, and the torus is

$$T = \prod_i \mathrm{Res}_{k_i/k} \mathbf{G}_m.$$

We prove the vanishing needed for this torus. A torsor under $\mathrm{Res}_{k_i/k} \mathbf{G}_m$ has transition functions which are units in $k_i \otimes_k R$ on a splitting cover $\mathrm{Spec}\, R \to \mathrm{Spec}\, k$. The same functions glue a rank-one module over k_i, by faithfully flat module descent. Conversely a rank-one k_i-module has precisely these local frames and transition functions, so these two constructions are inverse. Every rank-one module over the field k_i has a basis. Its frame torsor has a section, hence is trivial. Taking the product gives $H^1(k, T) = 1$. This is the multiplicative descent statement often called Hilbert 90, with its proof and its restriction-of-scalars application included here.

The root-height filtration from the fourth lesson descends to the twisted unipotent radical U. Each successive quotient is a vector group with a semilinear Galois action. We also give the additive vanishing explicitly. Let L/k be a finite Galois extension splitting the data and the given cocycle. For a semilinear L-vector space V and an additive cocycle c_σ,

$$c_{\sigma\tau} = c_\sigma + \sigma(c_\tau).$$

Choose $x \in L$ with $\sum_\sigma \sigma(x) = 1$, and set $b = \sum_\sigma c_\sigma \sigma(x)$. Changing the index by $\sigma = \tau\rho$ gives

$$\tau(b) = \sum_\rho (c_{\tau\rho} - c_\tau)(\tau\rho)(x) = b - c_\tau.$$

Thus every such cocycle is a coboundary. The element x exists in every characteristic. Distinct field automorphisms are linearly independent as maps $L \to L$: in a shortest nonzero relation $\sum_j a_j \sigma_j(y) = 0$, normalize one coefficient to one, evaluate at zy, and subtract $\sigma_1(z)$ times the original relation. Choosing z on which another automorphism differs from σ_1 gives a nonzero shorter relation, a contradiction. Consequently the trace map $\sum_\sigma \sigma$ is nonzero. Its image lies in k, so scaling an element of nonzero trace gives trace one. This argument does not divide by the order of the Galois group.

For completeness, vanishing for the successive vector quotients implies vanishing for U as follows. Project a U-cocycle to the first vector quotient and use the preceding formula to make its image zero by a change of trivialization. A vector in this quotient lifts after enlarging the finite Galois splitting field: the root-coordinate product for the split filtration provides such a lift. The changed cocycle now lies in the next filtration subgroup. Repeat; the finite root-height filtration ends at the

identity. Hence $H^1(k, U) = 1$. First project a twisted Borel cocycle to T, whose torsor has already been proved trivial, and change its trivialization accordingly. The residual cocycle is under this U, so is trivial too. Thus $H^1(k, B_0^{\mathrm{ad}}) = 1$ for the Borel twisted by the fixed Γ-torsor.

There is therefore no further Borel-preserving inner twist. The Γ-class is recovered from the canonical quotient of the full automorphism group, so it is independent of the chosen Borel. The pinning provides a section on cohomology, and the vanishing just proved leaves exactly that section as the quasi-split class. Thus the based-datum class determines the quasi-split group. □

Over a general base, the analogous vanishing can fail. For a line bundle L on S, the adjoint group $\mathrm{PGL}(\mathcal{O}_S \oplus L)$ has the Borel stabilizing $\mathcal{O}_S$ and a diagonal torus. Its two root lines are L and L^{-1}, in one order or the other. A pinning for this Borel pair requires a frame of the positive root line. If L is nontrivial, that frame does not exist. The field proof used the vanishing of the Picard group of a field; it cannot be transplanted to an arbitrary scheme.

6. Inner forms of the special linear group

A **central simple algebra** over a field k is a finite-dimensional simple algebra with centre k. Its matrix form and the separability of its splitting field will be proved below. We first identify the automorphisms that carry its descent data.

Lemma 6.1. The automorphism functor of the matrix algebra is PGL_n, over any base ring. Thus PGL_n-torsors correspond to degree-n Azumaya algebras.

Proof. Let ϕ be an algebra automorphism of $\mathrm{Mat}_n(R)$. Its images $p_i = \phi(E_{ii})$ of the diagonal matrix units are orthogonal idempotents, and $R^n = \bigoplus_i p_i R^n$. Each summand is finitely generated projective, since it is the image of an idempotent. It has rank one on every fibre: after a residue-field extension, the corner $p_i \,\mathrm{Mat}_n\, p_i$ is the endomorphism algebra of the corresponding vector-space image, whereas ϕ identifies this corner with $E_{ii}\,\mathrm{Mat}_n\, E_{ii}$, of dimension one. If the image has dimension r_i, its corner has dimension r_i^2, so $r_i = 1$. Local freeness of finite projective modules, proved in the preceding algebra lesson, now makes these rank-one locally direct summands over R itself. Locally choose a generator v_1 of the first summand, and put $v_i = \phi(E_{i1})v_1$. The matrix-unit identities show that the v_i's are a basis and that $\phi(E_{ij})$ sends v_j to v_i and kills the other v_h's. Hence ϕ is conjugation by the basis matrix. Two such matrices differ by a scalar, because a matrix commuting with every matrix unit is scalar. This proves the functor identity, with local lifts and descent through $\mathrm{GL}_n \to \mathrm{PGL}_n$. Twisting the matrix algebra by a torsor gives its Azumaya algebra; conversely the sheaf of local algebra isomorphisms is that torsor. Faithfully flat descent of the algebra and its multiplication proves effectivity. □

Theorem 6.2 (separable matrix splitting). Let k be any field and let A be a finite-dimensional simple k-algebra with centre k. Then $\dim_k A = n^2$ for some n, and there is a finite separable extension K/k with $A \otimes_k K \simeq \mathrm{Mat}_n(K)$. Conversely, an algebra obtained by faithfully flat descent of Mat_n over a field is central simple. Its algebra-isomorphism torsor has group PGL_n.

Proof. We first prove the finite-dimensional density calculation needed here. Let R act on a finite-dimensional simple k-module V, and suppose $\mathrm{End}_R(V) = k$. For linearly independent $v_1, \dots, v_m$, the map

$$R \longrightarrow V^m, \qquad r \longmapsto (rv_1, \dots, rv_m)$$

is surjective. For $m = 1$ this is simplicity. Induct on m. The image $L \subset V^m$ projects onto V^{m-1}. Its kernel in the last summand is an R-submodule of V, hence is either V or zero. The first case gives surjectivity. In the second case L is the graph of an R-linear map $V^{m-1} \to V$. Each component of this map is an R-endomorphism of V, hence multiplication by a scalar c_i. Applying the graph equation to $r = 1$ gives $v_m = \sum_{i<m} c_i v_i$, a contradiction. Taking a basis of V proves that the action

map $R \to \mathrm{End}_k(V)$ is surjective.

Apply this calculation to $R = A \otimes_k A^{\mathrm{op}}$ acting on $V = A$ by $(a \otimes b)x = axb$. Its submodules are the two-sided ideals of A, so V is simple. An endomorphism commuting with left multiplication is $x \mapsto xc$, where c is its value at 1; commuting also with right multiplication forces $c \in Z(A) = k$. Density and equality of dimensions therefore give an isomorphism

$$A \otimes_k A^{\mathrm{op}} \xrightarrow{\sim} \mathrm{End}_k(A). \tag{6.2}$$

Tensoring this linear isomorphism with any field extension E/k gives the same isomorphism for A_E. A two-sided ideal of A_E is stable under every endomorphism of its underlying E-vector space, so is zero or all of A_E. Also $Z(A_E) = E$: centrality is the kernel of the finite family of linear commutator maps with a k-basis of A, and kernels commute with field extension. Thus central simplicity survives every field extension.

Now take $E = \overline{k}$ and a nonzero simple left A_E-module V, obtained by quotienting by a maximal left ideal. Its endomorphism algebra is a finite-dimensional division algebra over the algebraically closed field E. Every one of its elements is scalar: its characteristic polynomial as an E-linear operator has a root c, so the operator minus c is singular; in a division algebra this makes it zero. Density gives $A_E \to \mathrm{End}_E(V)$ surjective, and simplicity makes its kernel zero. Consequently $A_E \simeq \mathrm{Mat}_n(E)$, where $n = \dim_E V$, and $\dim_k A = n^2$.

To obtain a *separable* splitting, consider the scheme $I = \mathrm{Isom}_{k\text{-alg}}(A, \mathrm{Mat}_n)$. Choose bases of the two algebras. A linear isomorphism is a matrix with invertible determinant; preservation of 1 and of the products of basis vectors is a finite list of polynomial equations. Thus I is an affine scheme of finite presentation. Its base change to $\overline{k}$ has a point by the preceding paragraph. Composition with that point identifies the entire scheme $I_{\overline{k}}$ with the automorphism scheme of the matrix algebra, hence with $\mathrm{PGL}_{n,\overline{k}}$ by Lemma 6.1. With the quotient convention for projective bundles, PGL_n is the determinant-nonvanishing open in $\mathbf{P}(\mathrm{Mat}_n^\vee)$, which parameterizes lines in the matrix module: the determinant is a homogeneous section of degree n, and invertible matrices modulo a unit scalar have exactly these projective matrix classes. On a chart where a matrix entry is a unit, normalize that entry to one; this gives a local section of $\mathrm{GL}_n \to \mathrm{PGL}_n$, and its inverse image is the product of that chart with $\mathbf{G}_m$. Thus the open projective scheme represents the quotient on all test rings. It is smooth over every base because projective-space charts are affine spaces and smoothness is preserved under open restriction. In particular $I_{\overline{k}}$ is smooth. The faithful field-extension descent proof in Supporting group-scheme proofs, Lemma G.2.1, makes I/k smooth, without a perfectness or characteristic hypothesis.

A nonempty smooth finite-type scheme over k has a point over a finite separable extension. Here is the precise reason, including for imperfect k. Over a separable closure k^s, choose a nonempty open with an étale map to affine space, as supplied by smooth coordinates. Its image is open. The infinite field k^s has a rational point in every nonempty open of affine space: a nonzero polynomial cannot vanish on all its tuples, by induction on the number of variables. The nonempty étale fibre over this point has a closed point finite separable over k^s, hence rational because k^s is separably closed. All coordinates of this point and of the chosen charts lie in a finite separable subextension of k^s/k. Applying this to I supplies the asserted K and splitting.

Finally, for an algebra B that becomes a matrix algebra after a faithfully flat field extension, a nonzero proper two-sided ideal would remain a nonzero proper ideal after extension. Matrices have no such ideal: multiplying a nonzero entry on its two sides by matrix units produces all matrix units. Centrality again consists of linear equations, so $Z(B) \otimes E = Z(B_E) = E$ and $Z(B) = k$. The isomorphism sheaf is locally the automorphism group of matrices, and is therefore the PGL_n-torsor of Lemma 6.1. Effective descent of its algebra operations is precisely the affine descent already proved in *Quotients and torsors*. This proves both directions, without a perfectness or characteristic hypothesis. □

This is the finite-dimensional density method for central simple algebras, followed by smooth descent of their matrix-isomorphism scheme. Compare the Stacks authors' splitting-field theorem, which also proves existence of a separable splitting field. The argument above proves the precise input needed for inner forms; Tsen's theorem and a computation of a field's entire Brauer group are not needed here.

Over the finite separable splitting extension in Theorem 6.2, A becomes Mat_n. The determinant descends to its **reduced norm** $\mathrm{Nrd} : A^\times \to \mathbf{G}_m$. Put

$$\mathrm{SL}_1(A) = \ker(\mathrm{Nrd}). \tag{6.1}$$

Conjugation of matrices acts simultaneously on the algebra and on SL_n. Twisting both by the same torsor shows that every marked inner form is (6.1), and every such algebra gives one. The reduced norm is well defined because the determinant is invariant under conjugation, so its local definitions agree on overlaps. This proves the assertion in arbitrary characteristic.

For $n \geq 3$, transpose-inverse is the nontrivial diagram automorphism of SL_n. Its action on the algebra torsor replaces A by A^{op}. Indeed, transpose identifies matrix multiplication with the opposite multiplication, and conjugation by g becomes conjugation by g^{-T}. Thus, if inner markings are forgotten, the resulting algebras are identified by $A \leftrightarrow A^{\mathrm{op}}$. The underlying groups are isomorphic by $a \mapsto a^{-1}$, viewed as a homomorphism from $A^\times$ to $(A^{\mathrm{op}})^\times$. For $n = 2$ the diagram automorphism is inner; the marked classification is still by the degree-two algebras.

7. Flags and the general linear group

Let $n = n_1 + \cdots + n_r$, with all $n_i > 0$, be an ordered composition. The corresponding standard parabolic consists of invertible block upper triangular matrices with block sizes $n_1, \ldots, n_r$. It stabilizes the partial flag with dimensions

$$d_j = n_1 + \cdots + n_j \quad (1 \leq j < r).$$

The order matters: the subset of cuts in $\{1, \ldots, n-1\}$, not an unordered partition, is the parabolic type containing the fixed upper triangular Borel.

For an S-scheme S', let $\mathrm{Flag}_{d_1,\ldots,d_{r-1}}(S')$ consist of chains

$$0 \subset F_{d_1} \subset \cdots \subset F_{d_{r-1}} \subset \mathcal{O}_{S'}^n$$

of locally direct summands of the indicated ranks, with locally free successive quotients. Use the subbundle Grassmannians, obtained by dualizing the quotient Grassmannians of AG-RG-03 Lemma 5.A. In their product impose the vanishing of each bundle map $F_{d_j} \to \mathcal{O}^n / F_{d_{j+1}}$. In local frames its finitely many entries generate the ideal of a closed incidence subscheme. Its test points have exactly the desired inclusions. The quotient $F_{d_{j+1}}/F_{d_j}$ is automatically locally free: the exact sequence

$$0 \to F_{d_{j+1}}/F_{d_j} \to \mathcal{O}^n/F_{d_j} \to \mathcal{O}^n/F_{d_{j+1}} \to 0$$

splits locally because its last term is locally free. This verifies the incidence functor over arbitrary rings. The product of the closed Plücker embeddings is projective, so its closed incidence subscheme is projective too.

The open chart where $F_{d_j} \to \mathcal{O}^{d_j}$ is an isomorphism for every j, using projection to the first d_j coordinates, is represented by block lower unitriangular matrices with diagonal blocks the identity. Unit-pivot elimination gives the matrix uniquely from a flag and conversely sends its first blocks to

that flag; it only inverts the indicated invertible projection determinants. Its free blocks give affine space of dimension

$$\sum_{i<j} n_i n_j. \tag{7.1}$$

The corresponding charts for coordinate permutations cover the scheme. On a residue-field flag, row-pivot elimination chooses nested coordinate sets I_j, of sizes d_j, for which all those projections are isomorphisms; one permutation sends them to the initial sets. Their determinant conditions define open charts on the original scheme, and coverage of all residue-field points is coverage of its underlying space. Thus these actual affine charts prove smoothness over S, including when S is nonreduced. Finally every flag admits an adapted basis Zariski locally: successively split its locally free quotients and choose bases of the summands.

Hence GL_n acts transitively as a sheaf, with stabilizer the block upper triangular P. It follows directly on the functor of points that

$$\mathrm{Flag}_{d_1,\dots,d_{r-1}} \simeq \mathrm{GL}_n / P. \tag{7.2}$$

Taking every cut gives the full flag scheme, representing Borel subgroups; its relative dimension is $n(n-1)/2$. Omitting every cut gives the one-point scheme representing the whole group as a parabolic. This is the linear-algebra model of Theorem 2.1.

Schubert cells for split flag schemes

Theorem 7.1 (relative Schubert cells). Let G/S be pinned split of constant datum, let $J \subset \Delta$, and let P_J be its standard parabolic. Let W_J be the Weyl subgroup generated by J, and W^J the minimal-length representatives for the right cosets W/W_J. Then G/P_J has a finite disjoint stratification by locally closed subschemes

$$U_{\Psi_w} \longrightarrow G/P_J, \qquad u \longmapsto u n_w P_J, \qquad w \in W^J,$$

each isomorphic to $\mathbf{A}_S^{\ell(w)}$. These cells commute with arbitrary base change. In particular the theorem applies over $\mathbf{Z}$ and supplies both a torification by coordinate tori and, over a finite field, the count $|(G/P_J)(\mathbf{F}_q)| = \sum_{w \in W^J} q^{\ell(w)}$.

Proof. Let U_J^- have the negative roots outside the root subsystem generated by J. The relative cocharacter open-cell theorem identifies $U_J^- \times P_J$ with an open subscheme of G. Quotienting this open set by its right P_J-action gives an open immersion $U_J^- \to G/P_J$. To justify the quotient assertion, its inverse image in the P_J-torsor $G \to G/P_J$ is the indicated right-stable open product, and the quotient of that product is U_J^-. Openness and the isomorphism descend through this faithfully flat torsor.

Minimality of w means $w(\Phi_J^+) \subset \Phi^+$. Thus $w^{-1}\Psi_w$ is negative and contains no Levi root: a negative Levi root $-\beta$ would give the positive root $w(-\beta)$, contradicting that condition. The ordered negative-root product makes $n_w^{-1} U_{\Psi_w} n_w$ a closed subscheme of U_J^-, by setting the other coordinates to zero. Translating the previous open chart by n_w proves that the displayed map is a locally closed immersion. Its source is the product of $\ell(w)$ framed root lines, hence affine space of that dimension over S.

For coverage on a geometric fibre, write $v = wz$, with $w \in W^J$, $z \in W_J$ and $\ell(v) = \ell(w) + \ell(z)$. Then $\Psi_v = \Psi_w \sqcup w\Psi_z$. The arbitrary-order root-product theorem of AG-RG-04 Section 5, with the first set first, gives

$$U_{\Psi_v} = U_{\Psi_w}\, n_w U_{\Psi_z} n_w^{-1}$$

as a product of schemes. The second factor and n_z are absorbed on the right in P_J, so the image of the Bruhat cell for v is the displayed cell for w. Field Bruhat coverage therefore covers every geometric point of the quotient.

Each displayed cell is the entire B-orbit of $n_w P_J$. Indeed the decomposition $U^+ = U_{\Psi_w} U_{\Theta_w}$ of AG-RG-04 Section 5 has $w^{-1}\Theta_w \subset \Phi^+$, so U_{Θ_w} fixes this coset, as does T. We also verify the needed indexing of T-fixed geometric cosets. If gP_J is fixed, $g^{-1}Tg \subset P_J$. The field conjugacy theorem for maximal tori of the connected smooth affine P_J supplies $p \in P_J$ with $gp \in N_G(T)$. Hence every fixed coset is $n_v P_J$. Two representatives give the same coset exactly when their Weyl elements differ on the right by the image of $N_G(T) \cap P_J$, which is W_J by the Bruhat description in Theorem 1.1. Thus the fixed cosets are precisely W/W_J.

Choose an integral cocharacter positive on every positive root. On $U_{\Psi_w} \simeq \mathbf{A}^{\ell(w)}$ its root coordinates have strictly positive weights, so its fixed point, and consequently its only T-fixed point, is the origin $n_w P_J$. If two displayed cells met on a geometric fibre, their entire B-orbits would coincide; their unique T-fixed points would then coincide, contradicting the distinct cosets in W/W_J. The images are therefore pairwise disjoint. Every point of the relative quotient admits a geometric point above it, so this gives disjoint coverage of the underlying space over S. The locally closed subschemes themselves were constructed over S before this fibre argument.

Every map, root-coordinate closed immersion and open chart used here commutes with base change, as does G/P_J by Theorem 2.1. Thus the locally closed stratification commutes with every base change, including nonflat changes to nonreduced schemes. Over $\mathbf{F}_q$, each rational point belongs to exactly one cell, and its affine coordinates give $q^{\ell(w)}$ points; summing gives the stated count. Over $\mathbf{Z}$, partition each affine cell coordinate by zero versus invertible. The resulting locally closed pieces are split tori and partition geometric points and points over every field, giving the claimed torification. A stratification does not assert that the coproduct of its strata equals the original fppf sheaf: a point over a ring with nilpotents can cross a stratum boundary and need not factor through one stratum. All constructions here still represent their stated subfunctors on every scheme test. □

For $G = \mathrm{GL}_n$ and block type $(n_1, \dots, n_r)$ this applies to the flag scheme just constructed, with $W = S_n$ and $W_J = S_{n_1} \times \cdots \times S_{n_r}$. It retains the ordered block type and the locally direct-summand condition on flags. For a nonsplit form, these globally indexed affine cells are not asserted: étale-local splitting of its parameter scheme alone does not give such a global stratification.

8. Exercises and solutions

Exercise 1 *easy.* Show that $\tau(g) = (g^T)^{-1}$ is outer on SL_n for $n \geq 3$, and is conjugation by $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ on SL_2.

Solution. Reversing the order twice makes transpose-inverse a homomorphism; it is an involution. If it were inner by h, its preservation of the diagonal torus would put h in that torus's normalizer. Its action on $X^*(T) = \mathbf{Z}^n/\mathbf{Z}(1, \dots, 1)$ is -1, whereas the Weyl action is a coordinate permutation. For three distinct indices, a permutation carrying $e_i - e_j$ to $e_j - e_i$ for every pair would have to exchange both (i, j) and (i, k), which is impossible. Thus -1 is not a Weyl action for $n \geq 3$. For $n = 2$, direct multiplication, using $ad - bc = 1$, gives

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix} = \tau(g).$$

The identities hold over every ring, including characteristic two.

Exercise 2 *medium.* Classify the inner forms of SL_n over a field, specifying what changes when the inner structure is forgotten.

Solution. Its adjoint group is PGL_n. Lemma 6.1 sends a torsor to a degree-n central simple algebra A; applying that torsor to the determinant-one matrix group gives $\mathrm{SL}_1(A)$. Conversely the local splitting isomorphisms of A give the same torsor. This proves the classification with inner structure. For $n \geq 3$, forgetting it divides by the diagram involution, which sends A to A^{op}, as the transpose calculation in Section 6 shows. The inversion map supplies the corresponding group

isomorphism. The distinction concerns the marking; it does not create an additional family of inner groups.

Exercise 3 *medium.* Classify the parabolics of GL_n containing the upper triangular Borel.

Solution. The simple roots are $e_i - e_{i+1}$, for $1 \le i < n$. A subset J joins consecutive indices into blocks: put a cut at i precisely when $e_i - e_{i+1} \notin J$. The resulting ordered block sizes are positive and sum to n. The roots in the Levi are exactly differences within blocks; the other permitted root entries are above the blocks. Thus Theorem 1.1 gives exactly the block upper triangular groups. Recovering their cuts recovers J, so there is neither repetition nor omission. For $n = 3$, the four types are $(1,1,1), (1,2), (2,1), (3)$.

Exercise 4 *hard.* Prove that the Borel subgroup functor of GL_n is represented by a smooth projective flag scheme over an arbitrary base.

Solution. Form the closed incidence subscheme in $\prod_{d=1}^{n-1} \mathrm{Gr}(d,n)$ defined by $F_d \subset F_{d+1}$. Its test points are exactly full locally split flags. The Plücker embeddings and the incidence equations prove projectivity. Adapted bases give local lifts to GL_n; their transition matrices preserve the flag exactly when upper triangular, so this scheme represents GL_n / B. The lower unitriangular charts have $n(n-1)/2$ free coordinates and prove smoothness. By Theorem 2.1 a Borel is locally a conjugate of B, and its normalizer is B. Hence it determines exactly one quotient section, with descent on overlaps. This identifies the flag and Borel functors on all schemes, rather than merely on field-valued points.

Exercise 5 *medium.* Explain why the automorphism group of a split torus need not be of finite type, and verify the central automorphism (3.2).

Solution. A torus automorphism is an invertible integral change of its character basis, so the representing scheme is the disjoint union of copies of the base indexed by $\mathrm{GL}_r(\mathbf{Z})$. For $r \ge 2$ it has infinitely many components, since the matrices $\left(\begin{smallmatrix} 1 & m \\ 0 & 1 \end{smallmatrix}\right)$ are distinct. On GL_2, centrality of the determinant factor proves that (3.2) is a homomorphism. Its determinant is $\det(g)^{-1}$, so applying it twice returns g. On scalar matrices it sends zI to $z^{-1}I$, whereas every inner automorphism fixes them. Thus it is a nontrivial outer automorphism arising from the central lattice direction.

The course prerequisite guide records the exact supporting statements, their full proof routes, and the hypotheses needed in their applications.

References and exact prerequisite proofs

- Michel Demazure and Alexander Grothendieck, *Schémas en groupes*, freely accessible Gille–Polo re-edition of 13 October 2024: Exposé XXIV and Exposé XXVI, for comparison with the automorphism, form and parabolic arguments proved here.
- Brian Conrad, *Reductive group schemes*, §§5.2 and 7.1–7.2. The proofs here use the preceding lessons' root-coordinate and pinned classification results.
- Milne's freely accessible *Algebraic Groups*, version 2.00 (2015). This exact free author edition provides comparison material; its citations do not replace the programme proofs.

The preceding written supporting proofs are Algebra and sheaf cohomology, §§3–8, for the finite-cover, affine and projective calculations; Projective cohomology and smooth affine models, Proposition 3.2 and Proposition 5.2, for retaining smooth projective models and line bundles and for direct images commuting with every base change; and Supporting group-scheme proofs, Lemma G.2.1, for smoothness descent over any field. The projective quotient and scheme-theoretic normalizer arguments are AG-RG-03 §5, using the earlier written descent, Zariski Main and flat-quotient proofs. AG-RG-03 Lemma 5.A supplies the quotient Grassmannian charts and closed Plücker immersion, with subbundle charts obtained by dualizing. The current written *Quotients and torsors* proof supplies faithfully flat module and affine descent, Theorems 6.1–6.3 for torsors and étale local

sections, Theorem 7.1 for cocycles, §7.12 for their Galois convention, and Theorem 8.1 for frames and Hilbert 90. The current written normalizer Lie calculation is *Lie algebras and smoothness*, Theorem 2.9. The separable matrix-splitting argument needed for inner forms is Theorem 6.2 of this lesson itself; no Tsen theorem or computation of an entire Brauer group is consumed. The proofs above establish the remaining parabolic, automorphism, form, flag and Schubert assertions. This identifies the consumed proof chain without claiming that unrelated parts of the programme are proof-complete.